ADAGIO GROUP

Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Re: Review #3 Convex Perturbation Paper

Benjamin D. Summers <bsummers@adagiollc.com> Sun, Sep 13, 2026 at 3:32 PM
To: "ISMAIL-SUTTON, SARA R." <sara.r.ismail-sutton@durham.ac.uk>, Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Hi Ezequiel and Sara,

Thank you both for your reviews (Ezequiel's attached). The revised manuscript has been uploaded to SSRN (https://dx.doi.org/10.2139/ssrn.6812461). You both reviewed the same text without conferring and converged on the same three issues. Below are my responses to each point, in manuscript order, starting with the one that explains most of the others. Where you both raised the same point, I answer it once.

Ezequiel, thank you for the verification record: the two-threshold point is the one I'd defend, and it stands unchanged.

The one that explains most of the others is what the convex combination is claimed to model. I had presented the convex combination as the canonical model of a state-conditioned interaction, and then let that reading carry the Discussion, which is exactly the inconsistency identified in Ezequiel's Finding 4 and Sara's item 29: the abstract disclaimed any interpretive weight for "observer coupling" while Section 7.5 relied on exactly that interpretation. Neither review found a realization theorem because there is none. Ezequiel's objection settles the question: a random choice between two dynamics, made once and conditioned later on the state, gives a weight that Bayes updates with the state and the time, not a constant. The revision presents the mixture as the simplest fixed-weight convex perturbation of an indivisible family, a formal path in the convex set of two-time families. The families are abstract two-time transition families in the sense of the Koopman paper's Definition 2.3; process-induced families are the special case; and every theorem is a statement about families, so nothing is lost by the restriction. "Canonical" and "observer coupling" are removed throughout, g is called the mixing weight, and the interpretive sentences in Section 7.5 are withdrawn.

The second change that runs through the paper is a piece of new mathematics that answers several points at once. With $\Delta = (1-g)[X - gK]$, where X is the substrate deficit operator and K the interaction operator, the deficit up to scale δt is exactly $(1-g)S(g)$, where $S(g) = \sup\|X - gK\|$ over the triples is convex and Lipschitz with constant M_* (Proposition 3.2). This result is used three times below: it states exactly when the mixed family is divisible at an interior coupling, it separates the rise of the bound from the behavior of the exact deficit, and it identifies the invariant object that the sign-blind bounds summarize.

1. Abstract (Sara 1, 2). Accepted. The abstract is rewritten at about half the length around the identity, the pencil form, the two thresholds, and the bounds, with one sentence for the secondary results. The statement "arbitrarily strong suppression at a computable sub-unit coupling" is replaced by the correct one: for each fixed ε the sufficient coupling is below 1 and tends to 1 as $\varepsilon \to 0$, and exact vanishing is guaranteed only at $g = 1$.
2. Introduction (Sara 5, 6, 7, 8, 9, 10, 11). Accepted as a group. The interaction term is named in the sentence that introduces the residual-mass scaling. The claim that the quantitative consequences are "not elementary" is removed; the techniques are a bilinear expansion, the triangle inequality, convexity of a norm, and compactness, and the claims are stated at that level. The introduction no longer asserts a transfer to non-Markovianity functionals; Remark 7.1 says "may induce" and identifies the missing continuity or comparison inequality. The sentence stating that the theorem "specializes to CPTP perturbations of process tensors" is deleted; it was unproved and contradicted Section 7.2, which now states that no process-tensor analogue or specialization is claimed. The paragraphs on time reversal, process tensors, and speed limits are removed from the introduction and replaced by one paragraph on the closest literature (item 13). The results imported from the Koopman paper are listed in one sentence; everything else is stated as proved here.
3. Section 2, conventions (Sara 12, 13, 14; Ezequiel on the no-generator statement). Accepted. Probability vectors are columns, $p_t = \Gamma(t \leftarrow s)p_s$, and the text states this explicitly against the row-stochastic convention. D_{CK} is described as the deficit up to temporal scale δt, the worst case over durations at most δt; the symbol is unchanged. Lemma 2.3 no longer "extends" anything; the same formula is continuous on the closed set. The no-generator statement is scoped as Ezequiel asked and as revision 4 of the Koopman paper states it: no time-ordered-exponential representation in the strongly measurable, Bochner-integrable generator class on the observed state space, which excludes an ordinary time-local evolution family and nothing else. It appears once, in Section 2, as orientation; this answers Question 5. Remark 2.4 now states that the results concern the two-time CK deficit only and do not address, or claim to eliminate, memory that is invisible at the two-time level (Sara 8).
4. Section 2, the model (Ezequiel Finding 3 and Question 1; Sara 3, 4, 15). Accepted; this is the change described above. Remark 2.7 states three things the paper does not claim: that Γ_g is the transition family of a process with a compatible hierarchy of joint laws; that any protocol produces the mixture with a time-independent weight; and any interpretation of g beyond a mixing weight. Question 1 is answered by the second of the two

alternatives Ezequiel offered: no realization is required, and the paper says why. Remark 6.2 gives the branch-mixture calculation. A Bernoulli(g) choice between processes Y and Z produces the kernel of Section 6.1 with weight $g \cdot \pi^{Y}_{s}(j) / [g \cdot \pi^{Y}_{s}(j) + (1-g) \cdot \pi^{Z}_{s}(j)]$, which depends on s and on the state and reduces to the constant g only when the one-time laws of Y and Z coincide. The constant-weight model is therefore the marginal-matched special case of the state-dependent family, and Proposition 6.1 is the estimate that applies to the general branch mixture. I verified the formula numerically against direct Bayes computation on random kernels, including the marginal-matched case.

5. Section 3, the identity and its consequences (Sara 16, 17). Accepted. The orders are stated in the correct variable: the interaction term is second order in the substrate–reference distance $W = \|\Gamma_0 - \Gamma_{ref}\|$ on the two subintervals, and the substrate deficit is first order in itself. Remark 3.3 separates two claims I had combined: $M > D_0$ is the condition for the bound to rise near $g = 0$, and it says nothing by itself about the exact deficit, whose behavior is determined by the alignment of K with X. When $K = cX$, the exact deficit is $(1-g)|1 - cg|\|X\|$, which for $c > 1$ falls to zero at $g = 1/c$ and then rises, and for $c \le 1$ is monotone. An initial rise of the exact deficit requires $M > D_0$ and an interaction term directed against X. Xu and Jagadish are cited here for non-additivity under mixing.
6. Section 7.1, "that is, at g = 1" (Ezequiel Finding 2; Sara 27). Accepted. The clause was introduced in the July revision, when I corrected the Buscemi–Datta overclaim that the terminal threshold restores information monotonicity; the correction replaced an overclaim with a false uniqueness inference, and the paper's own example refutes it. The correct statement is Proposition 3.2(iii): for $g < 1$, the mixed family is divisible up to scale δt if and only if $X = gK$ at every triple, so if the substrate is indivisible there is at most one interior divisible coupling, and there it equals $\|X\|/\|K\|$. In the interaction-dominant example $K = cX$ with $c \approx 2.78$, so the interior zero is $1/c \approx 0.360$ (0.3603 at full precision, the value Ezequiel computed); this answers Question 3. Remark 7.1 now states that exact monotonicity holds where the family is divisible, which is guaranteed at $g = 1$ and possible at most at one interior coupling, and that above g_{term} the deficit is suppressed to tolerance, not reduced to zero.
7. Section 4, the thresholds (Sara 18, 19; Ezequiel Question 7). Accepted. The definitions use strict inequalities, so a sentence after Definition 4.1 states that the infimum need not belong to the set and that the deficit may equal ε at a threshold; the theorem distinguishes behavior at a threshold from behavior strictly above it. Sara's quadratic envelope is now the main statement of Theorem 4.3, $g_{term} \le g_1 = 1 - [D + M - \sqrt{((D+M)^2 - 4M\varepsilon)}]/(2M)$, with the linear bound $1 - \varepsilon/(D+M)$ as a corollary within the same theorem. The proof shows $E > \varepsilon$ exactly on $[0, g_1)$, E decreasing on $[g_1, 1]$, and $g_1 \le g_0$. In the example, g_1 is 0.523 against 0.543 in the slow regime and 0.807 against 0.834 in the fast regime, against actual thresholds of 0.477 and 0.108. Ezequiel's Question 7 is answered here. M_* has no invariant meaning beyond being the supremum of $\|K\|$. The invariant object is $S(g)$, the norm of the pencil $X - gK$ maximized over triples, and the exact response is $(1-g)S(g)$. The bounds are what remains when the pencil is summarized by the two numbers D_{CK} and M_*, and that summary discards sign by construction; the factor-of-eight gap in the example is a consequence of the summary, not of slack in the argument. S itself is not a bound and cannot be sharpened. Remark 4.5 states this. I checked the quadratic bound on three thousand random parameter sets, the $M \to 0$ limit, and the pencil identities on random multi-triple families to machine precision.
8. Remark 4.6, mixing estimates (Sara 20). Accepted. The remark is deleted rather than repaired. Sara's two objections are both correct: the gap-only exponential bound requires a reversible or normal generator, and mixing of the reference alone does not bound M_*, which depends on the substrate–reference distance on both subintervals. Nothing in the paper used the remark. The Mitrophanov, Fill, and Szehr–Reeb–Wolf references are removed with it.
9. Section 4.2 and the product form (Sara 21, 28; Ezequiel Question 8). Accepted. Remark 4.11, the rate scale, is deleted. Proposition 4.8 states the stability result as a uniform lower bound $\eta(\varepsilon)$ on g_{first} over a band of scales, and the product form is reduced to one sentence, which identifies it as a restatement with no independent content. Section 7.4 is one paragraph: the resemblance to speed-limit inequalities is formal only, and the rate-form question posed in the earlier version is withdrawn.
10. The two-state example (Sara 22). Accepted with one reservation. The example is defined at one triple, and the text now states this at the start, replaces the scale-level quantities by their pointwise counterparts explicitly, and no longer says "verbatim." I did not extend it to a complete family. The two-state family of the Koopman paper's Section 6, mixed with the same Glauber reference, would provide one, and the text says so; but the purpose of the example is to exhibit the identity, the interior zero, the rebound, and the two bounds in a short calculation, and a full family would add a page without adding information. If either of you considers that the pointwise version undersells the scale-level theorems, I will add the family.
11. Section 5 (Sara 23, 24). Accepted in part. The Minkowski form is now the theorem, $\sqrt{J(\Gamma_g)} \le (1-g)\sqrt{J(\Gamma_0)} + g(1-g)\sqrt{J_M}$, with the earlier factor-of-two form noted as weaker; verified on random families. The retarded-operator material is compressed to one paragraph, since Theorem 5.1 uses only the pointwise deficit. I kept the section for one reason: J is the quantity the Koopman paper uses to characterize divisibility, and Theorem 5.1 is the one place where the two papers connect quantitatively. The introduction's causal-structure paragraph is removed.
12. Section 6 (Sara 25, 26). Accepted. The state-dependent construction is described as a pointwise construction that is a two-time family in the sense of Definition 2.1 only when the coupling function is continuous. Section 6.2 is retitled "Monotone reparametrization of the coupling" and states in its first sentence that it is a reparametrization of the same affine family, not a new coupling.
13. Priority (Ezequiel Finding 5 and Question 2; the Xu–Jagadish item). Accepted. Anand and Brun, De Santis and Giovannetti, and Xu and Jagadish are cited. The introduction's literature paragraph and the new Section 7.3 state that the terminal suppression threshold is a fixed-reference, tolerance-relaxed variant of the De Santis–

Giovannetti measure, and that the originality lies in the setting and the identity rather than in the idea of Markovian admixture. Ezequiel's four distinctions are confirmed as stated and are now stated against those references: the reference is fixed and the threshold is in the weight against it, not a minimization over admixtures; the object is the classical two-time CK deficit in operator norm, not CP-divisibility; suppression is to tolerance ε up to a temporal scale, not exact Markovianization at all times; and the exact identity, with its asymmetric interaction term and the first-crossing/terminal distinction, has no counterpart when both mixed dynamics are general. I have nothing to add to the list and nothing to remove from it.

14. Section 7.5, the scope of detectability (Ezequiel Finding 4, Questions 4 and 6; Sara 29). Accepted. The section is rewritten as a scope statement. Detectability is deterministic and population-level. From finite data, D_CK is estimated with sampling error, and "below ε" becomes a statistical statement that requires an estimator, its distribution under the null hypothesis that the family is divisible at the chosen scale, and a calibration of ε against that distribution. The paper supplies none of these and now says so; this is my answer to Question 6. "Indistinguishable from a divisible family" is qualified throughout as indistinguishable according to the CK-deficit statistic at the chosen scale and tolerance. The observer language is withdrawn, as in item 4; this answers Question 4. Sara's point about subspaces is in the text: the deficit subspaces of the substrate are independent of g, and those of the mixed family are not, since Δ(Γ_g) is proportional to X − gK and its range moves with g.
15. There are two recommendations I did not adopt, both noted above: extending the example to a full family (item 10) and removing Section 5's operator framing rather than compressing it (item 11). Both are decisions about length, not correctness, and I will reverse either if you think the paper is worse for it. On the assessment that the mathematical depth is modest, I have no counterargument. Proposition 3.2, the quadratic bound, and the Minkowski form are the only additions I could justify, and I prefer a short, correct paper to a longer one.

Bibliography: Mitrophanov 2003 and 2005, Fill 1991, and Szehr–Reeb–Wolf 2015 are removed; Anand–Brun 2019, De Santis–Giovannetti 2021, and Xu–Jagadish 2026 are added; the year of Bhattacharya–Bhattacharya–Majumdar is corrected to 2021. The companion citations are updated: the foundations paper is cited under its new title, "The Physics of the Instant," and the Koopman paper as revision 4. I checked the numbering against the compiled file; every result this paper cites from it retains its number.

Let me know your thoughts.

Ben

Benjamin D. Summers
Managing Director
ADAGIO GROUP
MOBILE +1.713.775.1011 | **OFFICE** +1.305.433.1823 | **FAX** +1 239.932.5558
bsummers@adagiollc.com | adagio.capital

On Fri, Sep 11, 2026 at 3:04 PM ISMAIL-SUTTON, SARA R. <sara.r.ismail-sutton@durham.ac.uk> wrote:

> Dear Ben
>
> The convex perturbation paper report was attached here.
>
> Best regards
>
> Dr Sara Ismail-Sutton
>
> ---
>
> **From:** ISMAIL-SUTTON, SARA R. <sara.r.ismail-sutton@durham.ac.uk>
> **Sent:** 04 September 2026 17:52
> **To:** Benjamin D. Summers <bsummers@adagiollc.com>
> **Subject:** Review #2 Re: Koopman Spectral Theory Paper
>
> Hi Ben,
>
> Thank you for your detailed response to the Koopman review.
>
> I appreciate you setting out where you agreed with the comments, where you took a different view, and how you intend to respond to the remaining reviews.
>
> For clarity, I had already completed most of my review of the Convex Perturbation paper using the version available on SSRN by the time your email arrived. If the current post-examination revision differs materially from that version, please let me know.
>
> Please find attached the review.